Comparative Study of Object Tracking Algorithms: KCF, CSRT, and TLD

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Contents

1 Introduction

Under the rapid development of the low-altitude economy, the demand for UAV tracking modules continues to rise in both industry and research. Classical discriminative trackers such as KCF, CSRT, and TLD have already provided important technical foundations for practical target tracking. KCF, proposed by Henriques et al., relies on circulant matrices and the kernel trick, and uses the fast Fourier transform in the frequency domain to achieve very efficient computation[2]. CSRT, proposed by Lukežič et al., introduces channel reliability and spatial reliability so as to improve robustness while maintaining good accuracy[4]. TLD, proposed by Kalal et al., combines tracking, learning, and detection, and adapts to target appearance variation through online updating[3]. However, each of these algorithms has an obvious shortcoming. KCF is extremely fast and has low computational cost, but it can easily lose the target after short-term occlusion. CSRT has higher precision, stronger robustness in complicated scenes, and better scale adaptation, but its computational cost increases significantly as its precision improves. TLD has persistent learning capability and good resistance to occlusion, but its heavy computational burden makes real-time performance difficult to maintain[3].

Although many studies have focused on UAV tracking, most existing work concentrates on improving a single algorithm rather than carrying out a systematic comparison of several classical algorithms under a unified benchmark. In particular, in UAV tracking, the target often undergoes drastic viewpoint change, the platform shakes frequently, and the available computational resources are limited. Therefore, the trade-off among accuracy, speed, and computational cost still requires more explicit analysis. Based on this background, this paper systematically compares KCF, CSRT, and TLD on standard datasets, quantitatively analyses their application boundaries in UAV tracking, and attempts to provide a clearer basis for algorithm selection and later optimisation.

At the same time, this project further proposes a hybrid KCF–TLD method. The aim is not simply to force two algorithms to run together, but to combine their advantages in a role-based way: KCF is used as the main tracker because of its speed, while TLD is used as a repair module when the target is lost. In this way, the study not only compares the three classical algorithms, but also explores whether a complementary design can achieve a more practical balance between high frame rate and target recovery capability.

2 Literature Review

According to the core mechanism of modern discriminative trackers, the algorithms discussed in this paper can be roughly divided into three categories: kernel correlation filter methods represented by KCF, channel and spatial reliability methods represented by CSRT, and long-term tracking-learning-detection methods represented by TLD[1].

2.1 KCF

The core idea of KCF is to train a discriminative classifier through a kernelized correlation filter. It generates a large set of translated samples by cyclic shifts of the target patch and learns the model by ridge regression. Its core frequency-domain solution can be written as

$$\hat{w} = \frac{\hat{x}^* \odot \hat{y}}{\hat{x}^* \odot \hat{x} + \lambda},$$

where $\hat{x}$ denotes the Fourier transform of the sample, $\hat{y}$ is the desired Gaussian response, and λ is the regularisation parameter. By diagonalising circulant matrices through the discrete Fourier transform, KCF greatly simplifies computation and therefore achieves a very high frame rate[2]. Its main strengths are speed and relatively low computational demand. However, its weaknesses are also clear: it is prone to bounding-box drift, its ability to handle large scale change is limited, and it usually loses the target after severe occlusion.

2.2 CSRT

CSRT is based on channel and spatial reliability. Its general optimisation idea can be expressed as

$$\min_{f} \sum_{c=1}^{C} \alpha_c \left\| S_c(f) * g - y \right\|^2 + \lambda \left\| f \right\|^2,$$

where f is the filter, g is the target template, y is the desired response, $S_c(\cdot)$ represents spatial reliability, and α_c denotes the channel reliability weight[4]. Its major advantage is better robustness and higher tracking precision, especially in the presence of appearance variation and background interference. Its weakness is the high computational cost caused by richer feature modelling and reliability estimation.

Table 1: Matlab code, with a Gaussian kernel (Core KCF logic)

Inputs

x: training image patch, $m \times n \times c$
y: regression target, Gaussian-shaped, $m \times n$
z: test image patch, $m \times n \times c$

Output

responses: detection score for each location, $m \times n$

```
function alphaf = train(x, y, sigma, lambda)
        k = kernel_correlation(x, x, sigma);
        alphaf = fft2(y) ./ (fft2(k) + lambda);
end

function responses = detect(alphaf, x, z, sigma)
        k = kernel_correlation(z, x, sigma);
        responses = real(ifft2(alphaf .* fft2(k)));
end

function k = kernel_correlation(x1, x2, sigma)
        c = ifft2(sum(conj(fft2(x1)) .* fft2(x2), 3));
        d = x1(:)' * x1(:) + x2(:)' * x2(:) - 2 * c;
        k = exp(-1 / sigma^2 * abs(d) / numel(d));
end
```

2.3 TLD

TLD contains three modules: tracking, learning, and detection. The tracking module is commonly implemented using Median-Flow to estimate short-term motion and detect tracking failure; the detection module scans the full image by sliding windows and cascaded classifiers; and the learning module updates the detector online using the P-N learning mechanism[3]. Its advantage is strong long-term recovery ability and better resistance to target disappearance or occlusion. Its disadvantages are high computational cost, strong dependence on successful initialisation, and unstable performance under fast motion or severe disturbance.

2.4 KCF–TLD

In view of the strengths and weaknesses of the above algorithms, this project proposes a KCF–TLD method. Since KCF is much faster than the other two algorithms, KCF is selected as the main tracker. To address the problem that KCF easily loses the target, TLD is used as an auxiliary recovery module. At the beginning of tracking, the target is initialised once; after the target is considered lost, TLD is activated to perform relocalization, and the recovered target box is then used to reinitialise KCF.

3 Methodology

3.1 Proposed KCF–TLD Algorithm

The proposed method is described below.

> Initialise KCF with the first frame and initial bounding box. For each subsequent frame, first attempt KCF tracking. If the KCF result is reliable, output it and continue. Otherwise, increase the KCF failure counter. Once the failure counter reaches the activation threshold, enter recovery mode and initialise TLD. During recovery mode, call TLD only at scheduled intervals. Reject recovery candidates that are too far from the last valid target state. Accept a TLD recovery only after repeated and mutually consistent detections. Then reinitialise KCF using the recovered box and return to normal tracking mode. If recovery persists beyond the maximum allowed duration, terminate recovery mode and reset the failure state.

The algorithm spends most of its runtime on high-speed KCF tracking. TLD is called only when KCF becomes unstable or loses the target, so that the hybrid method can improve the success rate while avoiding the heavy cost of running TLD continuously.

3.2 KCF as the Base Tracker

As the base tracker, KCF uses circulant matrices and kernel ridge regression to achieve efficient computation in the frequency domain. Given a base sample x, cyclic shifts of the sample form the training set, and the regression target y is Gaussian-shaped. In each frame, the location with the maximum response is taken as the new target position. This makes KCF highly suitable for real-time UAV tracking.

3.3 TLD as the Recovery Tracker

As the recovery algorithm, TLD is responsible for relocalization after target loss. A recovery candidate is accepted only when it satisfies overlap constraints with the last valid target box and passes continuous consistency checks. After the recovery box is accepted, KCF is reinitialized with the corrected box and the system returns to normal high-speed tracking.

3.4 Fusion Strategy

This method does not force KCF and TLD to contribute equally on every frame. Instead, the two algorithms are assigned different roles according to their strengths. KCF is responsible for primary tracking, while TLD is responsible for relocalization after failure. Therefore, the main objective of this design is not simply to make every frame more accurate, but to increase the probability of finding the target again once it has been lost.

3.4.1 Design Logic

The design is based on the complementarity of KCF and TLD. KCF is extremely fast and performs well when frame-to-frame target change is limited. However, once severe occlusion, disappearance, drift, or strong scale change occurs, KCF is often unable to recover by itself. TLD, although significantly slower, has explicit redetection capability. Therefore, the proposed method aims to retain the high FPS advantage of KCF while using TLD as a corrective mechanism when KCF fails.

3.4.2 Triggering TLD

The most important design issue is how to determine that KCF is no longer reliable. In the current implementation, TLD is triggered when the number of consecutive KCF failures or abnormal outputs reaches 2. An output is considered unreliable under two conditions. First, the IoU between the current box and the previous valid box is below 0.15. Second, the area ratio between the current box and the previous valid box falls outside the interval from 0.5 to 2.0. These thresholds are intended to detect obvious spatial discontinuity and sudden scale instability.

3.4.3 Controlling TLD Overhead

Since TLD can significantly reduce overall FPS, the algorithm limits how long and how often TLD is allowed to run. In recovery mode, TLD is not executed on every frame, but only once every 3 frames. In addition, the recovery state is protected by a timeout mechanism: the maximum duration of a single recovery phase is 120 frames. Once this limit is exceeded, the recovery state is forcibly terminated so that the system cannot remain trapped in slow recovery indefinitely.

3.4.4 Validating TLD Recovery

The algorithm applies three constraints before accepting a TLD recovery box. First, the candidate must have an IoU of at least 0.3 with the last valid target box. Second, the recovery is not accepted immediately after one detection; instead, it must be confirmed over 2 consecutive detections. Third, the two consecutive candidates must also be sufficiently consistent with each other, with IoU not below 0.45. These checks are designed to reduce false recovery and improve the reliability of relocalization.

4 Results and Analysis

4.1 Performance Metrics

This paper evaluates the algorithms mainly through IoU, success, precision, and FPS. At least three figures and tables should be provided in the final version, including a success curve, a precision curve, and an aggregate comparison table.

4.2 Algorithm Comparison

The experimental part of the paper should objectively compare KCF, CSRT, TLD, and the proposed KCF–TLD method, with particular emphasis on the proposed method. The purpose is to show, under a unified benchmark, how different tracking principles lead to clear differences in speed, resource demand, robustness, and recovery capability.

5 Discussion

The discussion focuses on why each algorithm behaves differently. KCF is attractive because of its speed, but it is weak after target loss. CSRT is usually more stable, but slower and more computationally expensive. TLD has explicit recovery capability, but its heavy computation reduces real-time performance. The KCF–TLD design therefore attempts to combine the speed of KCF with the relocalization ability of TLD in a role-based way rather than a full fusion way.

6 Conclusion

This paper proposes a possible way to combine KCF and TLD for UAV visual tracking. The main objective is to solve the problem that KCF cannot recover after target loss. In the proposed method, KCF is used during the normal stage to maintain a high frame rate, while TLD is triggered when KCF is considered unstable or has lost the target. Through continuous confirmation and threshold constraints, the method attempts to improve the reliability of recovery.

Experimental comparison is intended to show that KCF–TLD can improve KCF in terms of IoU, success, precision, and robustness while trying to preserve the high frame rate characteristic of KCF. However, the method still has clear limitations. Since the normal stage is still dominated by KCF, the upper performance bound remains constrained by KCF itself. In addition, TLD recovery is not always stable, so it cannot be treated as a completely reliable module. Future work may improve the instability judgement, strengthen recovery validation, and introduce stronger intelligent models to verify whether the recovered box is consistent with motion direction and short-term scale variation.

References

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